Asif Kamal Turzo

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Research Interests

Empirical Software Engineering, Peer Code Review, Machine Learning, Code Review Automation, Software Security, Program Analysis, Mining Software Repositories, Natural Language Processing, Human Aspects of Software Engineering.

Education

Wayne State University, Michigan, USA

Doctor of Philosophy

Department of Computer Science, August 2024

Current GPA: 3.98/4.0

Advisor: Dr. Amiangshu Bosu

Dissertation title: Broadening Participation in Open Source Software Communities with Newcomers' Onboarding and Retention.

Wayne State University, Michigan, USA

Masters of Science

Major in Computer Science, August 2022

Advisor: Dr. Amiangshu Bosu

GPA: 4.0/4.0

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

Bachelor of Science

Computer Science and Engineering, September 2017

GPA: 3.45/4.0

Publications

Journal Papers

*In the Software Engineering field, TSE, EMSE, and TOSEM are considered top-tier journals.

1. **(TSE-25)** **Asif Kamal Turzo**, Sayma Sultana, and Amiangshu Bosu. "From First Patch to Long-Term Contributor: Evaluating Onboarding Recommendations for OSS Newcomers." *IEEE Transactions on Software Engineering (TSE)*, vol. TBD, 2025, **Impact Factor: 9.2**

2. **(EMSE-24)** **Asif Kamal Turzo**, and Amiangshu Bosu, "What Makes a Code Review Useful to OpenDev Developers? An Empirical Investigation." *Empirical Software Engineering (EMSE)*, vol. TBD, 2024. **Impact Factor: 8.7**

3. **(TOSEM-23)** Jaydeb Sarker, **Asif Kamal Turzo**, Ming Dong, and Amiangshu Bosu, "Automated Identification of Toxic Code Reviews Using ToxiCR.", *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 32(5), july 2023, **Impact Factor: 4.6**

4. **(EMSE-23)** Sayma Sultana, **Asif Kamal Turzo**, and Amiangshu Bosu. "Code reviews in open source projects: how do gender biases affect participation and outcomes?." *Empirical Software Engineering (EMSE)* 28, no. 4 (2023): 92, **Impact Factor: 8.7**

Conference Papers

In the Software Engineering field, ICSE, FSE, and ASE are considered top-tier (A) conferences.

5. **(FSE-25)** Jaydeb Sarker, **Asif Kamal Turzo**, and Amiangshu Bosu. The Landscape of Toxicity: An Empirical Investigation of Antisocial Behaviors on GitHub. In *Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE 2025)*, vol. TBD, 2025. **Acceptance rate = (24.5%)**.

6. **(ESEM-23)** **Asif Kamal Turzo**, Fahim Faysal, Ovi Poddar, Jaydeb Sarker, Anindya Iqbal, and Amiangshu Bosu. Towards automated classification of code review feedback to support analytics. In *Proceedings of the 17th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*, vol. TBD, 2023. **Acceptance rate = (29%)**.

7. **(ICSE-21)** Rajshakhar Paul, **Asif Kamal Turzo**, and Amiangshu Bosu, "Why security defects go unnoticed during code reviews? a case-control study of the chromium os project. In *Proceedings of the 43th International Conference on Software Engineering(ICSE)*, page 1373-1385. IEEE, 2021. **Acceptance rate = (22%)**.

8. **(APSEC-20)** Jaydeb Sarker, **Asif Kamal Turzo**, and Amiangshu Bosu. A benchmark study of the contemporary toxicity detectors on software engineering interactions. In *Proceedings of the 27th Asia-Pacific Software Engineering Conference (APSEC)*, pages 218–227. IEEE, 2020. **Acceptance rate = (37%)**.

Short Papers

9. **(MSR-25)** Costain Nachuma, Md Mosharaf Hossan, **Asif Kamal Turzo**, and Minhaz F. Zibran, "Decoding Dependency Risks: A Quantitative Study of Vulnerabilities in the Maven Ecosystem," In 2025 *IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR)*, pp. TBD, 2025.

10. **(ICSE-21)** Rajshakhar Paul, **Asif Kamal Turzo**, and Amiangshu Bosu, "A Dataset of Vulnerable Code Changes of the Chromium OS Project," *2021 IEEE/ACM 43rd International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)*, 2021, pp. 244-245.

11. **(ASE-22)** **Asif Kamal Turzo**, "Towards Improving Code Review Effectiveness Through Task Automation." *37th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2022 (Doctoral Symposium)

Academic Employment

Assistant Professor, September 2025 - Present
Department of Computer & Information Science
University of Massachusetts Dartmouth, Dartmouth, MA

Postdoctoral Fellow, August 2024 - August 2025

Department of Computer Science
Idaho State University, Pocatello, ID
Advisor: Dr. Minhaz Zibran and Dr. Farjana Eishita

Graduate Teaching Assistant, May 2022 - August 2024
Department of Computer Science
Wayne State University, Detroit, MI

Graduate Research Assistant, August 2019 - April 2022
Department of Computer Science
Wayne State University, Detroit, MI
Project: Worked on NSF grant #1850475 under the supervision of Dr. Amiangshu Bosu. The research focused on building an empirical understanding of useful code review and then utilizing the understanding to build a reviewer recommendation system for Open Source Software projects.

Grant Writing Experience

— ISU ISGP grant: Toward Effective Security Defect Fixing through Peer Code Reviews. Co-PI: **Asif Kamal Turzo**, (Award amount: $5,000).

— URSSI Fellowship: Benchmarking Code Review Automation Tools for Software Quality and Security. PI: **Asif Kamal Turzo**, (Proposed award amount: $23,000, Under submission).

Teaching Experience

Course Instructor (UMass Dartmouth, MA, USA)

— CIS 461: Formal Methods for Software Engineering: Fall 2025

— CIS 602: Formal Methods for Software Engineering: Fall 2025

Course Instructor (Wayne State University, MI, USA)

— CSC 3110: Algorithm Design and Analysis: Spring/Summer 2024

— CSC 4500: Introduction to Theoretical Computer Science: Spring/Summer 2022

Teaching Assistant (Wayne State University, MI, USA)

— CSC 4996: Senior Capstone Project: Winter 2024

— CSC 4420: Computer Operating Systems: Spring/Summer 2023

— CSC 5991: ST: Scalable Microservices: Spring/Summer 2023

— CSC 4996: Senior Capstone Project: Fall 2023

— CSC 4996: Senior Capstone Project: Winter 2023

— CSC 4996: Senior Capstone Project: Fall 2022

Software Industry Experience

Senior Software Developer
EnamelBD Private Limited
From 11/2017 to 05/2018
Dhaka, Bangladesh

Software Developer
piSTRATEGY (Software and Consulting Company)
From 2/2019 to 6/2019
Dhaka, Bangladesh

Tools and Applications

Tool Name: Code Review Comment Classification Tool [GitHub]

— A tool developed in Python for classifying code review comments.

— Can be utilized to calculate code review analytics and to understand how effectively reviewers have identified software defects.

Project Name: VAT Checker Android Application

— Has saved government from 3 billion BDT(Bangladeshi Taka) VAT fraud.

— Has generated social awareness to prevent VAT fraud and achieved a number of national and international awards.

Awards and Achievements

— Outstanding Graduate Research Assistant Award in the Computer Science Department, Wayne State University.

— Champion at mBillionth Award, 2016 (Government and Citizen Engagement sector) for VAT Checker(it is considered one of the most prestigious awards in South Asia).

— Champion at National Mobile Application Award, 2016 for VAT Checker.

— Champion at BRAC Manthan Digital Innovation Award, 2016 (E-Governance and Institutions sector) for VAT Checker.

— NSF Travel Grant for attending ESEM-2023 conference.

— SIGSOFT CAPS award for attending ESEM-2023 conference.

Academic Services

Reviewer

- ACM Transactions on Software Engineering and Methodology (TOSEM) Journal: 2024-Present
- Software Quality Journal: 2022-Present
- Junior PC member at Mining Software Repository (MSR) 2022 Conference

Additional Reviewer

2021 IEEE/ACM 43rd International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)

Student Volunteer

- 43rd International Conference on Software Engineering (ICSE), 2021
- 37th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2022

Presentations

— "Towards Improving Code Review Effectiveness Through Task Automation" at ASE-2022 Doctoral Symposium, Michigan, USA

— "Towards Automated Classification of Code Review Feedback to Support Analytics" at ESEM-2023, Louisiana, USA

Reference

Available on request

April 28, 2026